

Smart Features Live Test — Auto Analysis

Generated on 2026-09-24 10:08 UTC

Executive Summary

This report summarizes the dataset quality, exploratory findings, and model performance for project "Smart Features Live Test".

Dataset Overview

File	sample_customer_churn.csv
Rows	300
Columns	12
Missing Values	0
Duplicate Rows	0
Cleaned	True

Modeling & Evaluation

Model Type	decision_tree
Problem Type	classification
Target Column	churn

Metrics

Metric	Value
accuracy	0.8333
precision	0.838
recall	0.8333
f1	0.8318
roc_auc	0.8281
confusion_matrix	{'labels': ['0', '1'], 'matrix': [[21, 7], [3, 29]]}

class_distribution	{'1': 32, '0': 28}
cv_scores	[0.7504347826086958, 0.8538467163960801, 0.6230434782608696, 0.75, 0.8071862348178138]
cv_mean	0.7569

Feature Importance

Feature	Importance
tenure_months	0.2306
satisfaction_score	0.1759
contract_type_Month-to-month	0.1407
monthly_charge	0.1279
support_tickets	0.1215
age	0.1171
signup_date_2024-08-04	0.0165
signup_date_2024-01-07	0.0159
signup_date_2024-06-19	0.0156
signup_date_2024-06-21	0.0121
signup_date_2024-06-12	0.0107
signup_date_2024-07-11	0.008
signup_date_2024-07-06	0.0074
total_spend	0.0
region_East	0.0
region_North	0.0
region_South	0.0
region_West	0.0
plan_Basic	0.0

plan_Premium	0.0
--------------	-----

AI Business Insight *AI-generated insight*

AI service is currently unavailable. This report contains verified analytical results only; AI-generated narrative could not be produced.

Limitations

Model predictions are estimates based on historical data and may contain uncertainty. AI-generated text explains verified analytical results but does not independently calculate metrics. Always validate significant business decisions with domain expertise.